

Umer Irfan

Systems Engineer · Software Engineering · Embedded Firmware · Applied Machine Learning

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PROFESSIONAL EXPERIENCE

AI/ML Engineer | Streaming Voice Agent

Aug 2024 to Mar 2025 · [Project](#)

Sales demo prototype under [Asif Mufti](#). ZEISS was a prospective customer.

- Built a **real-time sales voice agent** using Node.js, WebSockets and a Python backend on DigitalOcean. Combined WebRTC VAD, Whisper transcription and GPT-4 responses with configurable sales personas.
- Added PyAudio streaming, session handling and MySQL call records. Tuned buffering for spoken responses in **under four seconds**.

React Native Engineer | Project X

Nov 2024 to Jun 2025 · [Project](#)

- Built a smart-camera Android MVP with **device onboarding**, Firebase authentication, Google sign-in and notifications so users could pair cameras and receive alerts in one app.
- Worked on ESP32 camera firmware for OTA update fetching and MQTT telemetry.

DevOps Engineer | U.n.I Social App

May to Jul 2024 · [Website](#)

- Built **GitHub Actions deployment workflows** for container delivery to EC2. Configured NGINX routing to a higher-capacity instance, automating releases and handling load.

APPLIED AI AND RESEARCH

Kissan-Dost | Urdu Conversational Agriculture

2026 · [Project](#) · [Paper](#)

- Co-developed an **Urdu conversational AI interface** that gave farmers field-grounded advice over WhatsApp voice and text. Built a multi-hop ESP-NOW sensing mesh and ESP32 gateway.
- Two 45-day deployments supported a five-person pilot with near-daily chatbot use and irrigation decisions informed by its advice. Co-author at IEEE DCOSS-IoT 2026.

PARDA | Audio Privacy for Smart Glasses

2027 · [Project](#) · [Manuscript](#)

- Applied **SEAL-style redaction** (adversarial sensitive-content anonymisation) to conversation context. Distilled a retrieval-aware adversary and anonymizer into a **2B model** using 717 CANDOR conversations.
- On 143 held-out conversations, model-judged privacy leakage fell from 0.502 to 0.378 while utility rose from 0.875 to 0.900. First author. Submitted to IEEE PerCom 2027.

LLM-Enhanced Wearables | Health Guidance

2026 · [Project](#) · [Paper](#)

- Developed a **\$12.72 screenless wearable** and plain-language WhatsApp feedback. In a 20-person, 96-hour study, mean self-rated comprehension rose from 3.05 to 3.60 out of 5. Co-author.

WatchTower | Network Anomaly Detection

2025 · [Project](#) · [Report](#)

- Built a transformer detector with 28 packet features and PROSER open-set recognition. Optimized INT8 inference with TVM on Raspberry Pi 5. Reported **3.82 ms inference** and 93.66% accuracy.

SELECTED PRODUCT AND SOFTWARE PROJECTS

Wellness Lamp | React Native and ESP32

2026 · [Project](#) · [Source](#)

- Built a **React Native control app** for lighting themes, sound, BLE control and mood-based palettes. Synchronized ESP32 nodes so one app controlled multiple lamps.

Fintelligent | Client Website

2026 · [Project](#) · [Live site](#)

- Built a responsive public site in Next.js, React and TypeScript so prospective clients could find financing services and contact the firm.

Makers Lab Dashboard | Full-Stack Tool

2024 · [Project](#) · [Source](#)

- Built a **React, Express and PostgreSQL** dashboard that brought equipment, staff, tasks and activity into one lab workspace. Added role-based sessions and Docker Compose packaging.

CARDY | Interactive Hardware Challenge Platform

2023 · [Project](#) · [Firmware](#)

- Took an ESP8266 challenge platform from **concept to event deployment**. Designed the PCB, firmware and telemetry, then coordinated assembly and testing of 50 devices for **50 teams and about 200 participants**.

PixelPacker | Edge Neural Compression

2025 to 2026 · [Project](#) · [Writeup](#)

- Built a distilled INT8 CNN encoder with a **36-fold smaller memory footprint** and an Artix-7 acceleration path. Reported **over 10 times lower bitrate than JPEG** at comparable perceptual quality, enabling image transfer over constrained wildlife links.

For the Love of the Game | JPEG Decoder and ASCII Raycaster

2023 · [JPEG](#) · [Raycaster](#)

- Challenged myself to implement an **ISO/IEC 10918 JPEG decoder from scratch** in C++, covering bitstreams, Huffman decoding, inverse DCT and YCbCr-to-RGB conversion.
- Built an interactive **ASCII raycaster** in C++ with wall intersections, distance shading and real-time console rendering to understand a game engine end to end.

LEADERSHIP AND MANAGEMENT

Convener | LUMS Science Symposium

2024

- Led a week-long program for **550 participants** and managed **150 team members across seven departments**. Coordinated university approvals and sponsor outreach.
- Secured **PKR 200,000** in cash and in-kind sponsorships. Delivered **PKR 800,000 net profit** for the society.

Event Head | AI Nexus (SPADES)

2024 to 2025

- Selected **12 recruits from 300 applicants** and assigned team roles. Managed four months of preparation for a three-day competition round with **90 participants**.

TECHNICAL SKILLS

Applications and services

TypeScript · JavaScript · Python · React · React Native · Next.js · Node.js · Express · REST · WebSockets

Data and deployment

PostgreSQL · MySQL · Redis · Firebase · EC2 · Docker Compose · GitHub Actions · NGINX

Applied ML and audio

PyTorch · Whisper · WebRTC VAD · model distillation · retrieval · INT8 quantization · TVM

Connected applications

BLE · ESP32 · MQTT · OTA updates · Java · C · C++

EDUCATION

Lahore University of Management Sciences.

BSc Computer Science. Minor in Computer Engineering. Sept 2023 to Jul 2027. CS GPA 3.707/4.00.

HONOURS AND AWARDS

- 1st Place.** [Pakistan Semiconductor Summit Hackathon](#) 2026. RISC-V RTL verification.
- Dean’s Honour List.** LUMS. Three semesters.

- Best Project.** Intel ISEF Pakistan 2019.
- Best in Category.** Teknofest 2020. Gaziantep, Turkey.
- Team Pakistan.** [Young Physicists Tournament](#), Beijing (2018).